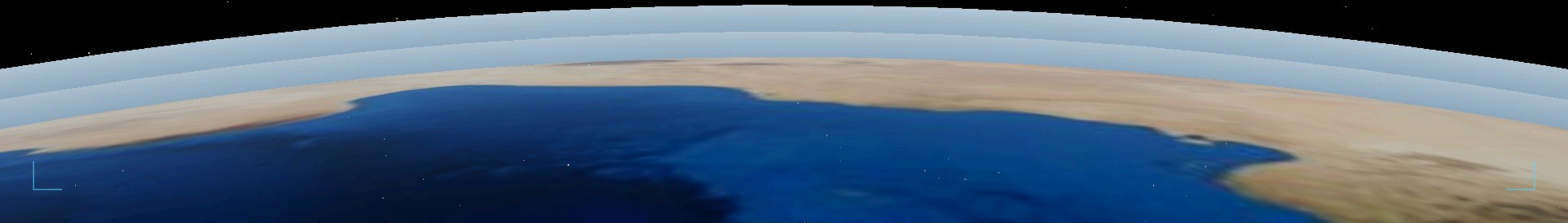
DECA INNOVATION PLAN · 2026–27

LOW EARTH ORBIT · 2026

ORBITAL CURBSIDE

The garbage truck for space.

One spacecraft. Ten pickups per trip. Earth does the steering.



INNOVATION PLAN

ORBITAL CURBSIDE

Cleaning up space junk ten pieces at a time, priced like a garbage route.



Brandon Graff
Founder & Chief Executive Officer

[School name]
[Street address] · [City, State ZIP]
[Presentation date]

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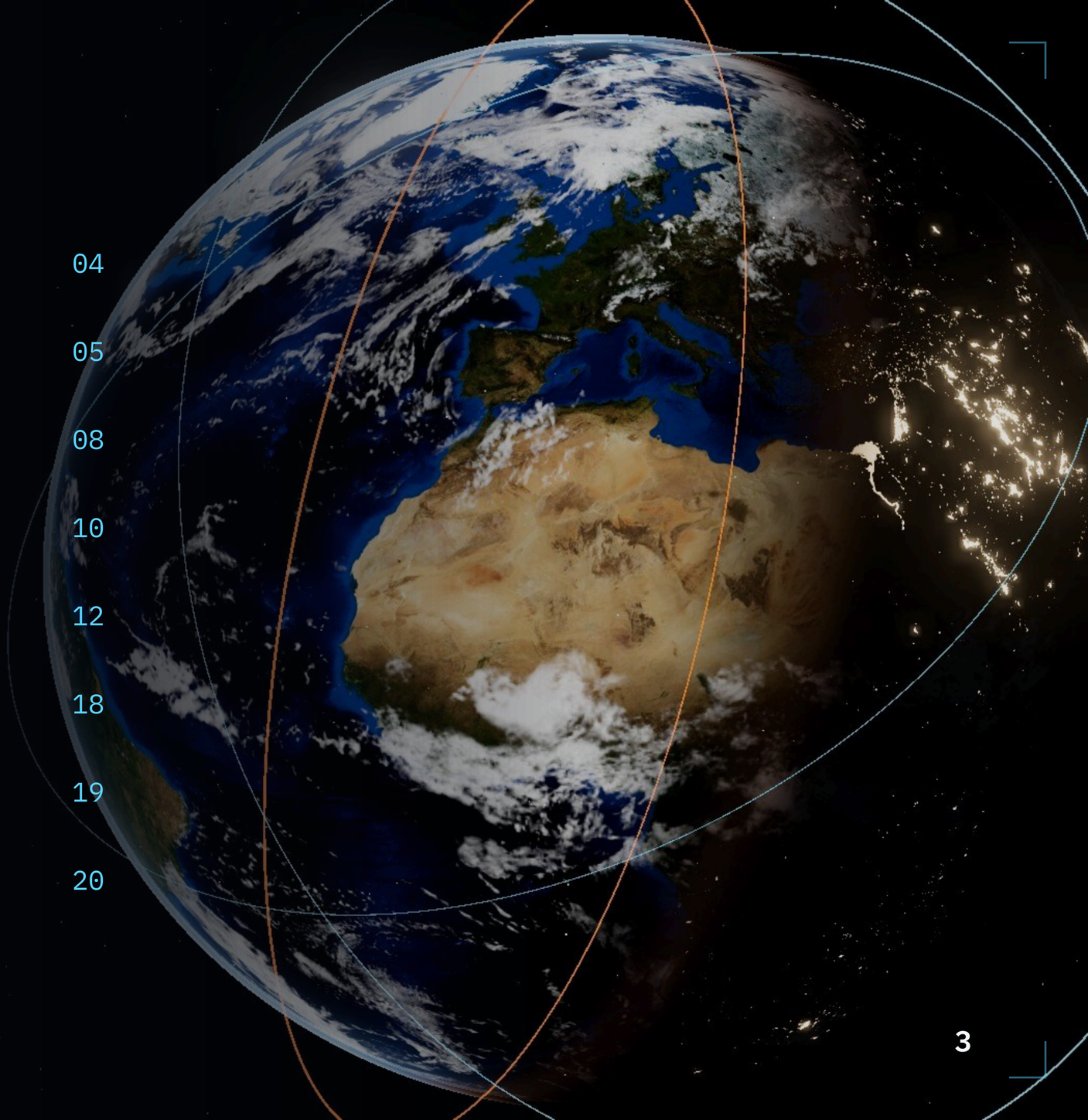
IV **Unique Value Proposition** 10
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SPACE HAS A TRASH PROBLEM. WE'RE BUILDING THE GARBAGE TRUCK.

THE PROBLEM

Dead satellites and old rocket parts are piling up around Earth, and every crash makes more junk.

OUR FIX

One spacecraft picks up **10 pieces per trip**, like a garbage truck on its route.

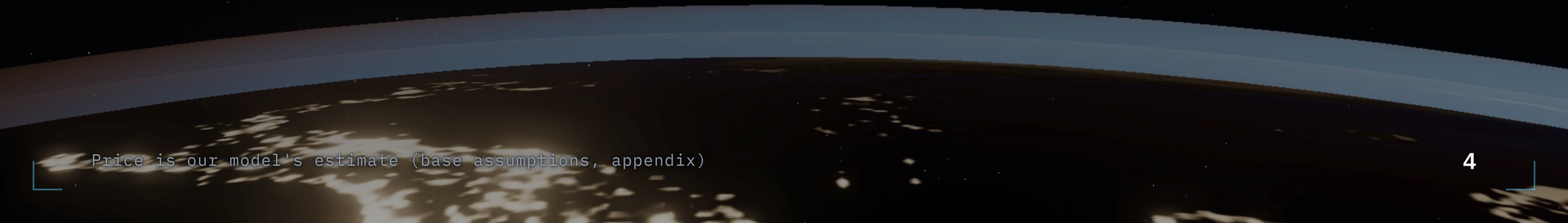
WHO PAYS

Space agencies first, then satellite companies and insurers. A fixed fee per piece removed.

THE PRICE

$12.2M

per piece in our model, launch included. Today: **$40–93M**.



Price is our model's estimate (base assumptions, appendix)

EARTH IS WRAPPED IN JUNK.

47,000
objects big enough to track are circling Earth

16,000
of them still work. The rest is junk.

27,000 km/h
how fast they travel (about 17,000 mph)

Picture a highway where two of every three cars are abandoned wrecks, still moving at full speed.



Every dot is a real object · blue: working · orange: junk

ESA DISCOSweb statistics (Oct 2026) · CelesTrak catalogue, 16 Oct 2026
speed = orbital speed at 400–1,000 km

WHEN JUNK CRASHES, IT MAKES MORE JUNK.

2009

A dead Russian satellite smashed into a working US one (Iridium 33). It created about **2,200** new pieces of tracked junk.

41×

times the International Space Station has had to move out of the way of junk since 1999. People live on board.

Chain reaction

Crashes now make new pieces faster than old ones fall. ESA says the junk keeps growing even if we stopped launching.

WHAT'S AT STAKE the satellites behind GPS maps, weather forecasts, TV and internet.

CelesTrak, Iridium 33/Cosmos 2251 collision · NASA Orbital Debris Quarterly News 29(3) · ESA Space Environment Report 2026

CLEANING UP COSTS $40–93 MILLION PER PIECE.

Every cleanup today is a custom mission for one or two objects. That's like hiring a whole garbage truck for every trash bag.

WHO PAID	PIECES	PRICE
Europe (ESA), 2020	1	€86M
Japan (JAXA), 2024	1	≈$82M
UK Space Agency, Oct 2026	2	≈$80M
US Space Development Agency, 2026	1 + options	$52.5M

AND THE RULES JUST GOT STRICTER

5 years — New US rule: dead satellites must come down within 5 years.

1 in 10 — Satellites allowed to fail at disposal. Some will always need a pickup.

$1.1–1.6B — Analysts' estimate of the debris market in 2025.

ESA; Astroscale; UK Government; Starfish Space · FCC 22-74 (2022), FCC 26-47 (2026) · Mordor Intelligence, ResearchAndMarkets

WHO PAYS TO TAKE OUT THE TRASH?

1

Space agencies

ESA · JAPAN · UK · US

WHY THEY CARE

Their old rockets are up there, and they already pay for cleanup.

HOW THEY BUY

Contracts, paid at milestones.

2

Satellite companies

IRIDIUM · ONEWEB · PLANET

WHY THEY CARE

Must bring dead satellites down within 5 years. Some will fail.

HOW THEY BUY

A fixed fee per satellite removed.

3

Insurers

SPACE UNDERWRITERS

WHY THEY CARE

They pay out when satellites are lost or hit.

HOW THEY BUY

Removal bundled with policies, once it's proven.

“I expect the cost is really in the millions or tens of millions, at which price I know it doesn't make sense.” — Matt Desch, CEO of Iridium (2019), on removing its dead satellites

Numbered by who pays first · Quote: C. Henry, SpaceNews, 30 Dec 2019

FIRST ROUTE: 10 OLD ROCKET STAGES FROM THE US, INDIA AND FRANCE.

4 India	4 France	2 USA

Where the cheap rockets go
Shared SpaceX launches drop satellites right in their orbit band, so our first pickup is 80 days after launch.

No permission problem
Their owners' own space agencies already pay for cleanup.

The bigger prize, later
142 Soviet-era stages. Russia must agree, and getting there takes a year, so about $13M each.



Our analysis of the CelesTrak catalogue, 16 Oct 2026 · Outer Space Treaty (1967), Art. VIII

ONE TRIP. TEN PICKUPS.

TODAY

One custom mission per piece



$40–93M
per piece

ORBITAL CURBSIDE

One truck, one route, ten stops



$12.2M
per piece, our model

Our promise: space cleanup priced like a garbage route. A fixed fee per piece, paid when it's confirmed coming down.

Our figure: model estimate with launch included, base assumptions (appendix) · Today's prices: slide 7

WE DON'T CHASE THE JUNK. WE LET EARTH BRING IT TO US.

Every orbit slowly swings around Earth, like the hand of a clock. Lower orbits swing faster, because Earth bulges at the equator.

So the truck parks a little higher or lower than its next target and waits until the two line up, like clock hands meeting.

TURN WITH FUEL
1,094 kg
of propellant

WAIT A YEAR
19 kg
of propellant



Fuel to close a 19° gap between two orbits, our model · J2 nodal precession (Vallado, 2013)

FOUR STEPS, TEN TIMES PER TRIP.



01

Ride along

The truck rides a shared SpaceX rocket, the cheapest seat to space.



02

Drift into line

It climbs to the junk's height and waits for Earth to line up the next target.



03

Grab and tag

A robot arm holds the old rocket. The truck latches on a small rocket kit.



04

Send it down

The kit fires. The old rocket falls and burns up. The truck drives to the next stop.

Renders from our 3D concept model, sized by our math. Not a flight design.

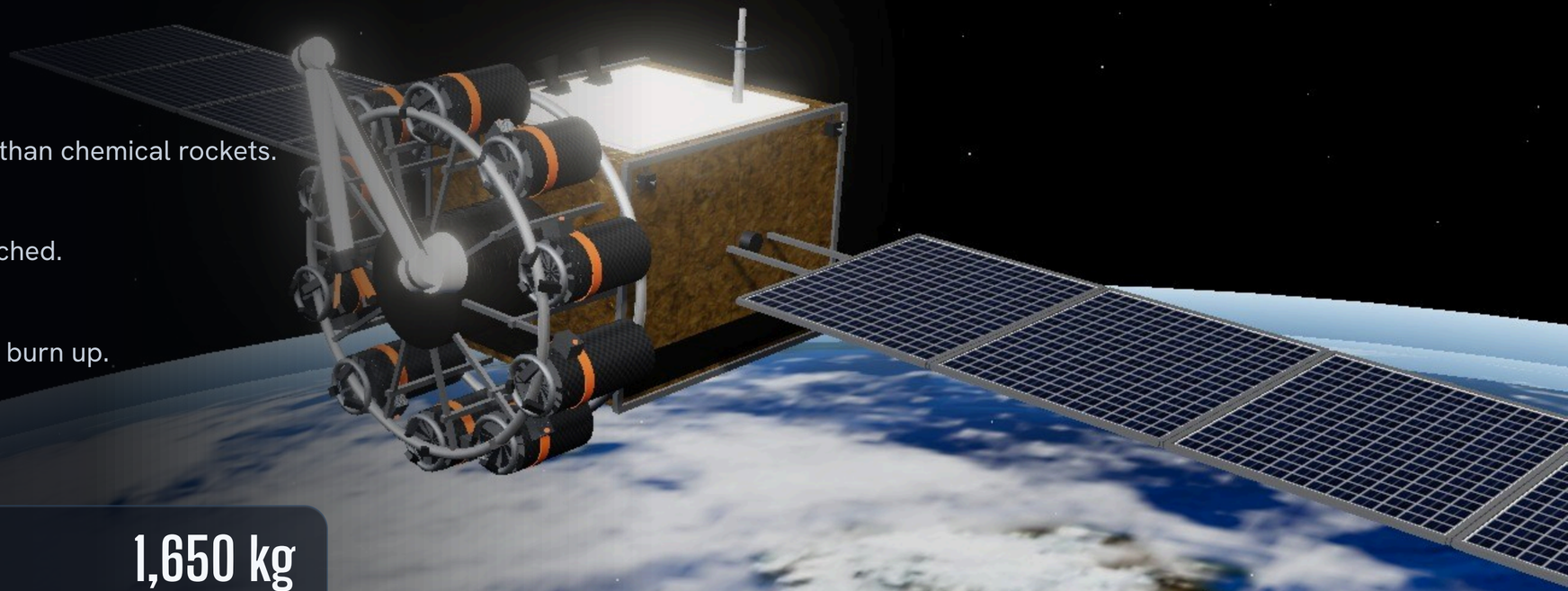
ABOUT AS HEAVY AS A CAR.

Electric engines
Run on xenon gas; about 5× more fuel-efficient than chemical rockets.

Robot arm
Holds each old rocket steady while a kit is attached.

10 rocket kits
One per pickup. Each sends one piece down to burn up.

Solar wings
Turn sunlight into power for the engines.



WEIGHT AT LAUNCH **1,650 kg**

Kits ×10 852 kg · Truck 600 kg · Xenon 198 kg

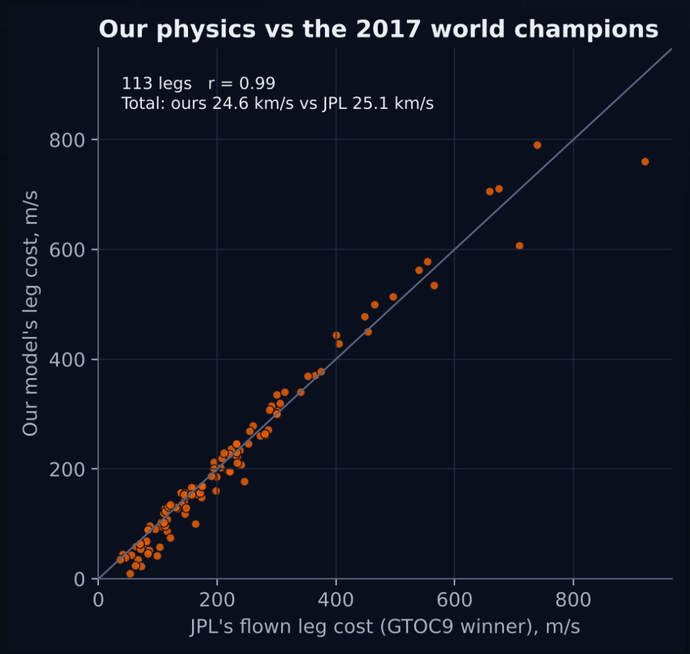
Concept sized from our model (first route). Not a flight design. Thruster class: Busek BHT-1500.

IS THE MATH RIGHT? WE CHECKED IT AGAINST NASA.

In 2017, the European Space Agency held a contest to plan routes like ours. NASA's Jet Propulsion Laboratory (JPL) won and published every move.

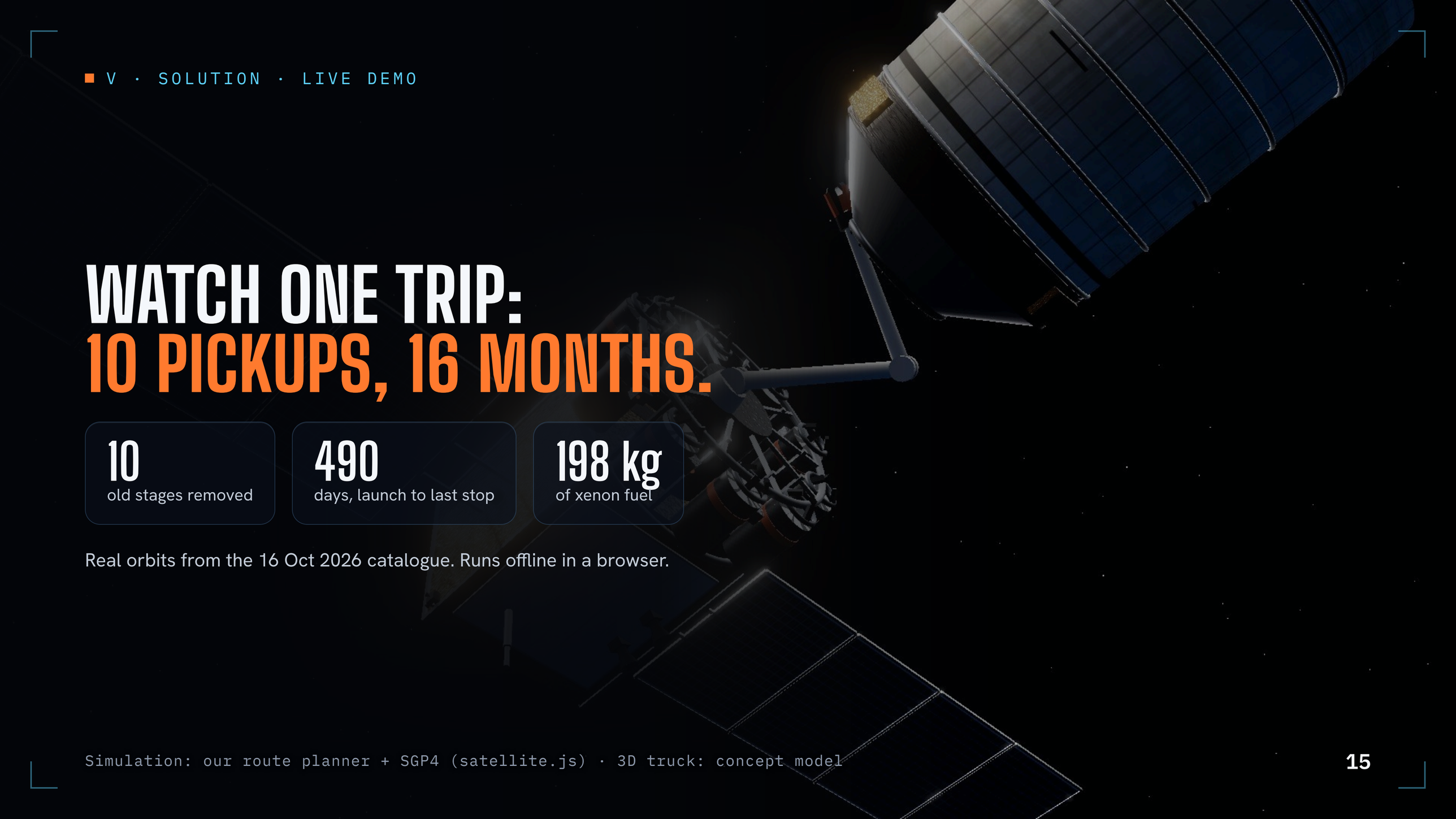
2.2%
Our fuel totals vs JPL across all 113 of their moves

0.06°
Our orbit predictions vs the model used to track real satellites (SGP4)



Each dot is one move from JPL's winning solution, recomputed by our model. On the line = perfect match.

Petropoulos et al. (2018), GTOC9 JPL results · Vallado et al. (2006), SGP4 · 0.30° on our first route



WATCH ONE TRIP:
10 PICKUPS, 16 MONTHS.

10	490	198 kg
old stages removed	days, launch to last stop	of xenon fuel

Real orbits from the 16 Oct 2026 catalogue. Runs offline in a browser.

Simulation: our route planner + SGP4 (satellite.js) · 3D truck: concept model

MORE PICKUPS PER TRIP, LOWER PRICE PER PIECE.



Cost per piece removed, launch included. 80% of 20,000 simulated cases land between $11.2M and $15.4M. **Biggest unknown:** what the truck itself costs ($50–115M).

Our model; inputs from NASA OTPS (2023), SpaceX rideshare prices, Busek · assumptions in appendix

WHAT COULD STOP US, AND WHAT WE DO ABOUT IT.

Grabbing it

Nobody has grabbed a tumbling old rocket yet. Japan plans to try in 2027–28; we'll partner with whoever proves it.

Cost

Truck and kit prices are estimates. I'm testing them with experts.

Permission

Every object belongs to its country. Our first route uses US, Indian and French stages.

Falling pieces

Heavy objects are steered to burn up over the open ocean.

2026–27
Talk to customers and experts; sharpen the model

2027–28
Sign a partner for the arm and spacecraft

2028–29
Small test trip: 2–3 pickups

2029+
First 10-pickup truck

Outer Space Treaty (1967), Art. VIII · Astroscale ADRAS-J2 · dates after 2027 are targets

LET'S TAKE OUT THE SPACE TRASH.

1. Space junk is growing, and every crash makes more.
2. Cleanup costs $40–93M per piece today. Our route: about $12M, launch included.
3. Earth does the steering, and the math is checked against NASA JPL.

THE ASK

[$__] in pre-seed funding for 12 months: finish customer interviews, design the rocket kit, and sign a test-trip partner.

Scan to fly the route

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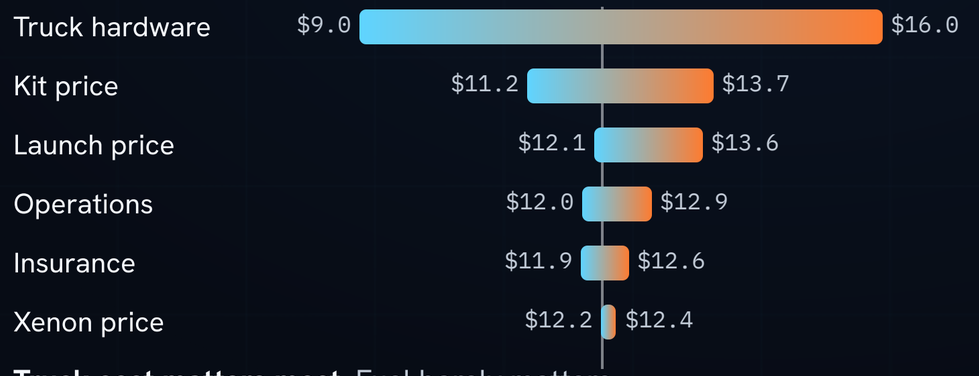
Earth imagery: NASA Visible Earth / Earth Observatory (Blue Marble, Black Marble, clouds). Renders: our own.

Full list with links: written paper, references [1]–[28]

INPUT	LOW · BASE · HIGH	SOURCE
Truck hardware	$50 · 80 · 115M	NASA OTPS (2023); base = assumption
Rocket kit, each	$0.5 · 1.5 · 3.0M	Assumption: no public price
Launch	$7.0 · 7.5 · 15k/kg	SpaceX rideshare to 550 km; truck climbs
Xenon	$2.8 · 2.8 · 12k/kg	Market reports (2024–26)
Operations	$3 · 5 · 10M/yr	Assumption
Insurance	5 · 8 · 12%	Assumption, of hardware + launch
Fuel margin	10 · 10 · 25%	Our JPL and SGP4 checks
Truck	600 kg dry; Hall thrusters, Isp 1,710 s; 10 days per stop	
Kits	Solid motor, Isp 285 s; lowers orbit to 250 km; ≈85 kg each (first route)	

WHAT MOVES THE PRICE MOST

Cost per piece if one input goes from low to high, others at base ($12.2M):



Truck cost matters most. Fuel barely matters.

Base case: first route, 10 pickups, 490 days from launch, launch included. Sources in paper.